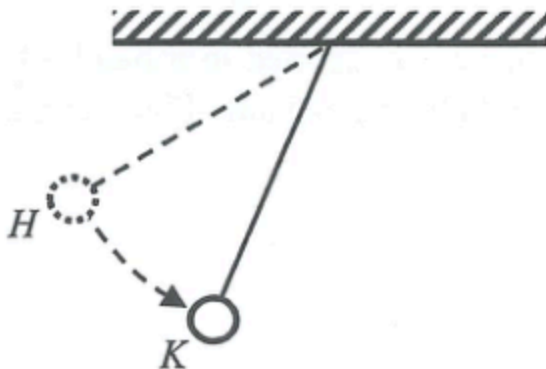


8.



A small sphere suspended by a light inextensible string is released from point H as shown. The string remains taut when the sphere swings downward. Which free-body diagram below best shows all the forces acting on the sphere at K ? Neglect air resistance.

A.



B.



C.

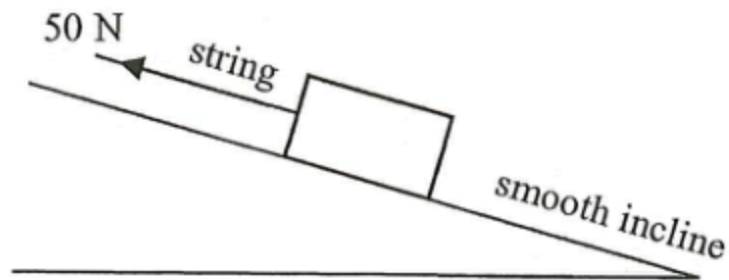


D.



9. A person thought that when he pulls his hair with a large enough force, eventually he will be able to pull himself off the ground. This is a misconception because
- A. the pulling force is acting internally within that person but it is not an external force acting on the person.
 - B. it is impossible to generate a pulling force larger than a person's weight.
 - C. the pulling force is balanced by the person's weight.
 - D. it is impossible for the pulling force to overcome the force due to atmospheric pressure.

6. On a smooth incline, a block of weight 120 N is kept stationary by a string with tension 50 N parallel to the incline.



The string is then cut and the block accelerates down the incline. What is the **net force acting on the block** before and after the string is cut ?

	before the string is cut	after the string is cut
A.	50 N	70 N
B.	50 N	50 N
C.	0 N	70 N
D.	0 N	50 N

5. A constant net force acting on an object of mass m_1 produces an acceleration a_1 while the same force acting on another object of mass m_2 produces an acceleration a_2 . If this net force acts on an object of mass (m_1+m_2) , what would be the acceleration produced ?

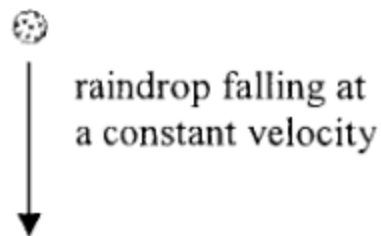
A. $a_1 + a_2$

B. $\frac{a_1 + a_2}{2}$

C. $\frac{a_1 a_2}{a_1 + a_2}$

D. $\frac{2a_1 a_2}{a_1 + a_2}$

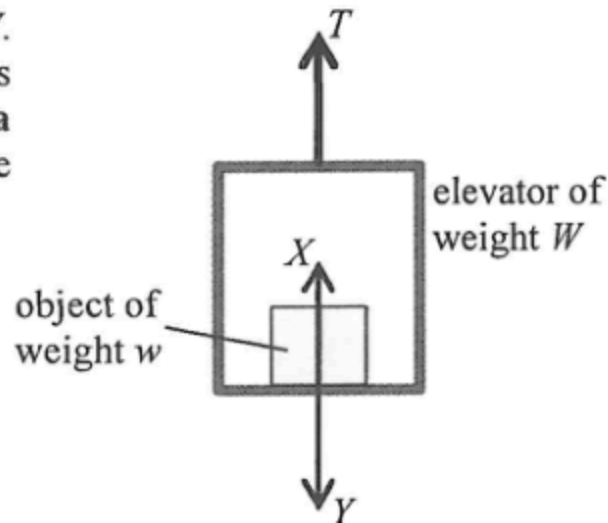
6. Which statement below about a raindrop falling at a constant terminal velocity is correct ?



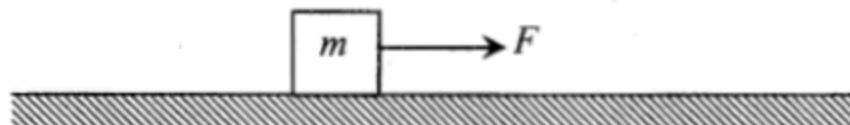
- A. No work is done on the raindrop by the gravitational force.
- B. As the raindrop falls, all its gravitational potential energy loss is converted into kinetic energy gain.
- C. The only force acting on the raindrop is its weight.
- D. No net force is acting on the raindrop.

6. An object of weight w is placed inside an elevator of weight W . A force T pulls the elevator upward. A **supporting force X** acts on the object while the **floor of the elevator experiences a force Y** due to the presence of the object. Which of the following is an action and reaction pair?

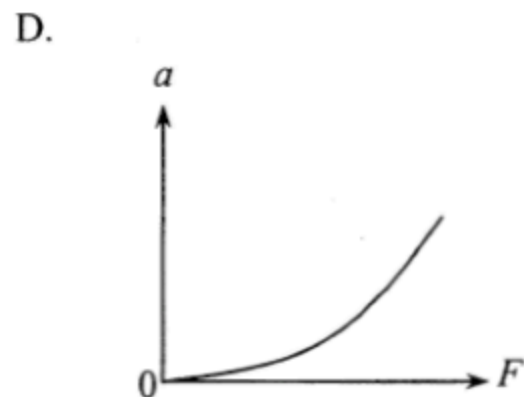
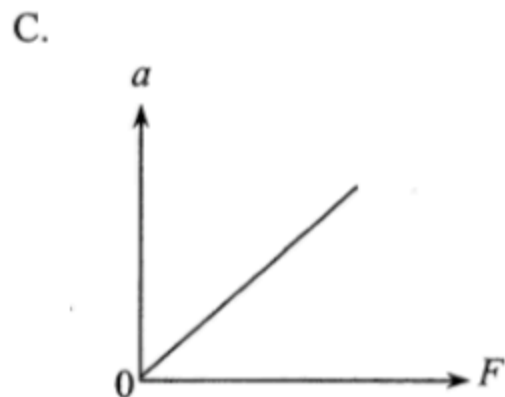
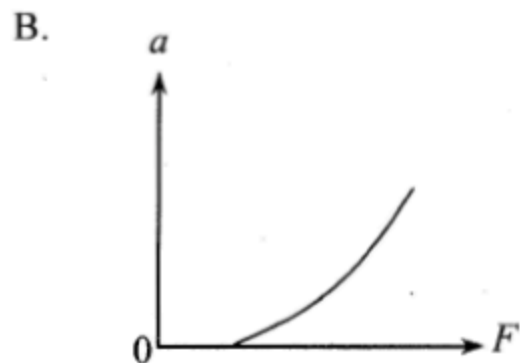
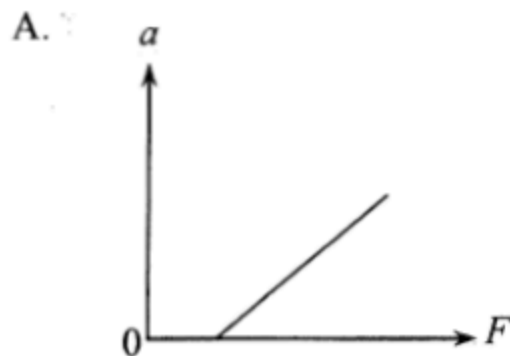
- A. T and $(W + w)$
- B. w and X
- C. X and Y
- D. It cannot be determined as the direction of acceleration of the elevator is unknown.



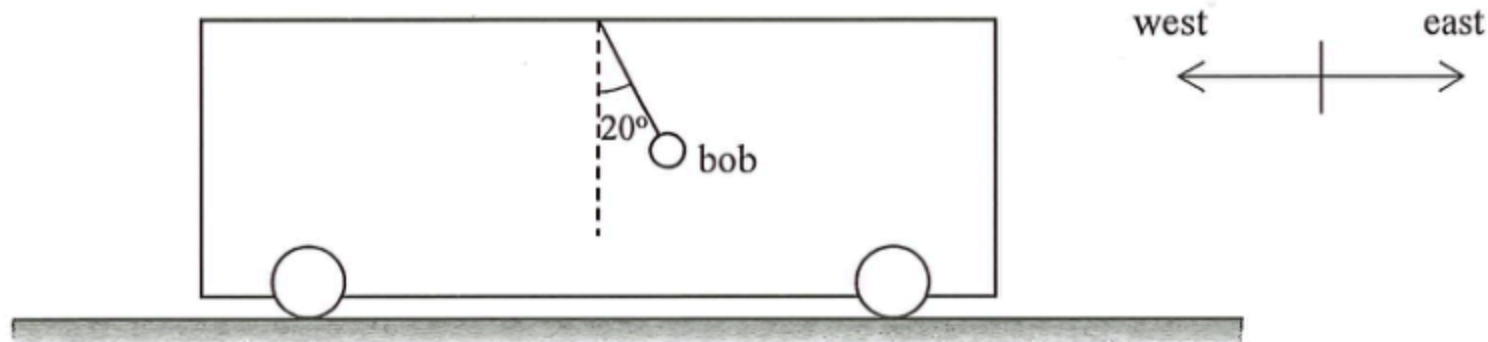
11.



A block of mass m initially resting on a rough horizontal surface is pulled along the surface by a horizontal force F increasing from zero. If the frictional force is constant, which graph shows the relation between the acceleration of the block a and force F ?



7. A bob is suspended from the ceiling of a train moving along a horizontal east-west rail.



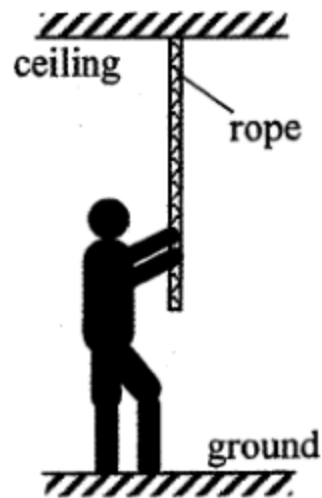
The bob slants towards east and makes an angle of 20° with the vertical as shown. What is the direction and magnitude of the acceleration of the train ? ($g = 9.81 \text{ m s}^{-2}$)

	direction	magnitude
A.	due west	3.36 m s^{-2}
B.	due west	3.57 m s^{-2}
C.	due east	3.36 m s^{-2}
D.	due east	3.57 m s^{-2}

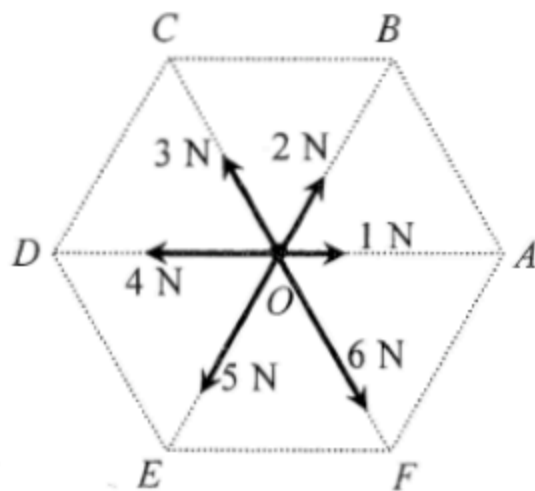
6. A boy of weight W exerts a downward pulling force F on a rope of weight G hung vertically from the ceiling. He stands still on the ground as shown. Which of the following gives the magnitude of the force exerted by

- (1) the boy on the ground;
(2) the rope on the ceiling ?

	(1)	(2)
A.	W	$G - F$
B.	W	$G + F$
C.	$W - F$	$G - F$
D.	$W - F$	$G + F$



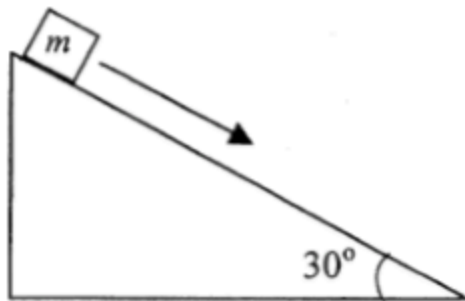
6.



In the figure, O is the centre of a regular hexagon. A particle at O is subject to six forces with magnitudes indicated as shown. The resultant force acting on the particle is

- A. 9 N along direction OE .
- B. 8 N along direction OE .
- C. 8 N along direction OF .
- D. 6 N along direction OE .

10. A block of mass m resting on a 30° incline is given a slight push and slides down the incline with a uniform speed. Which of the following statements about the block's motion on the incline is/are correct?



- (1) There is no net force acting on the block.
- (2) The frictional force acting on the block is $0.5\ mg$.
- (3) If the block is given a greater initial speed, it will slide down the incline with acceleration.

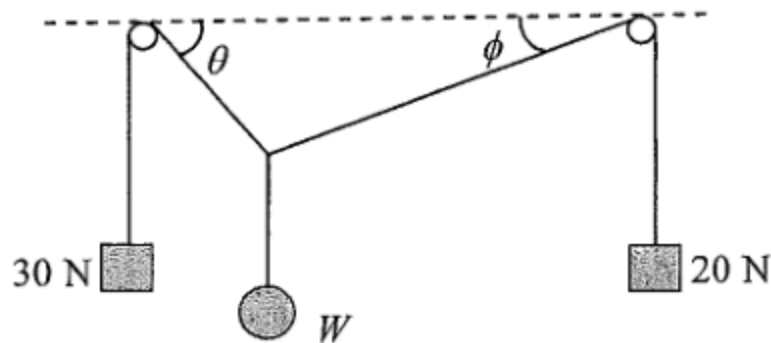
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only



Two forces F_1 and F_2 of constant magnitudes act at the same point as shown. When the angle θ between F_1 and F_2 increases from 0° to 180° , the magnitude of the resultant force

- A. decreases throughout.
- B. increases throughout.
- C. decreases and then increases.
- D. increases and then decreases.

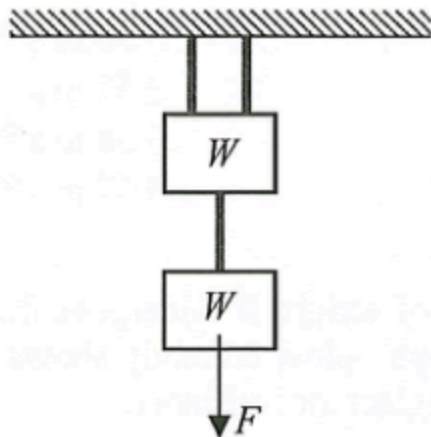
4.



The figure shows a weight W attached to two light strings which pass over two smooth pegs at the same height and with weights 30 N and 20 N attached to the respective ends of each string. The system is at equilibrium. Which deduction about W is correct ?

- A. W is less than 50 N.
- B. W is equal to 50 N.
- C. W is greater than 50 N.
- D. No information about W can be obtained as angles θ and ϕ are not known.

5. Two blocks, each of weight W , are hung by three identical light strings as shown. Each string has a limiting tension of $2W$, i.e. it will break for a tension larger than this value.

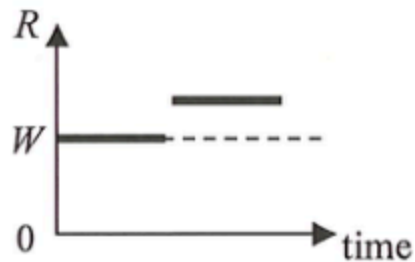


Force F acts vertically on the lower block. What is the largest value of F without breaking any of the strings ?

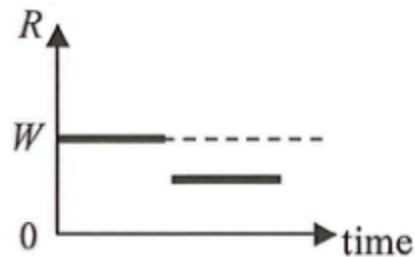
- A. $0.5W$
- B. W
- C. $1.5W$
- D. $3W$

8. In an amusement park ride, a boy of weight W undergoes free fall vertically in a chair at first and then decelerates uniformly. Which graph below correctly shows the time variation of the reaction force R acting on the boy by the chair? Neglect air resistance.

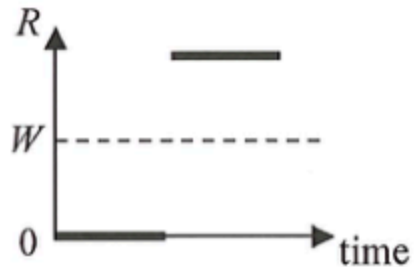
A.



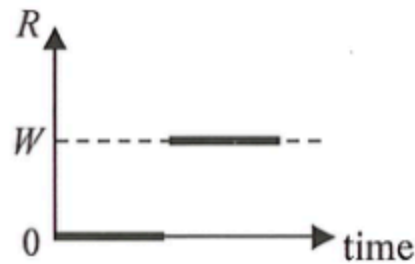
B.



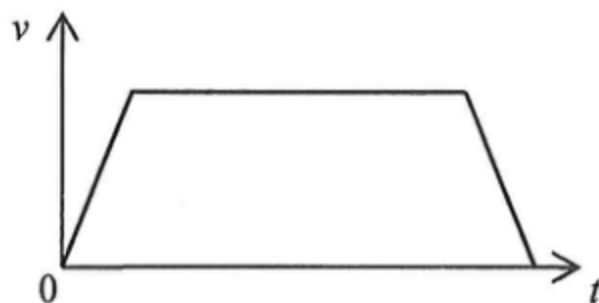
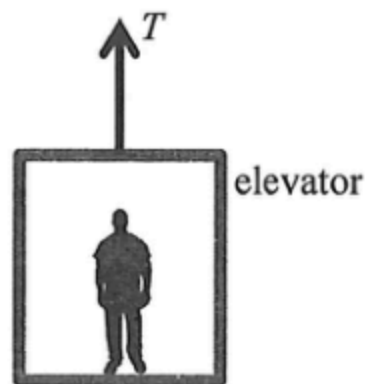
C.



D.

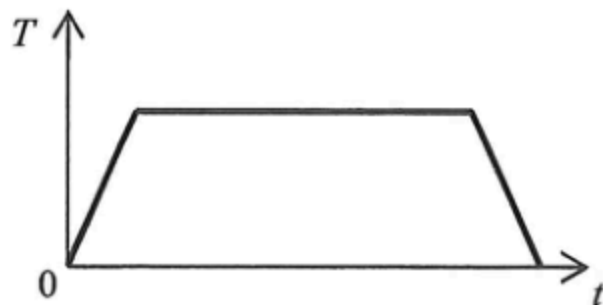


6.



An elevator is pulled upward by a cable. The v - t graph above shows the variation of the elevator's velocity with time. Which of the following correctly indicates the variation of the tension T in the cable with time t ?

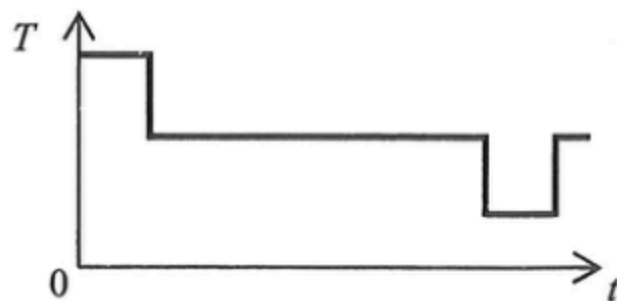
A.



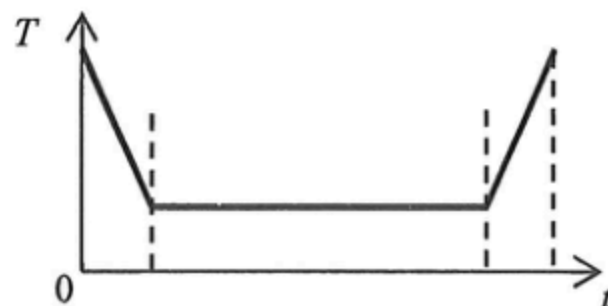
B.



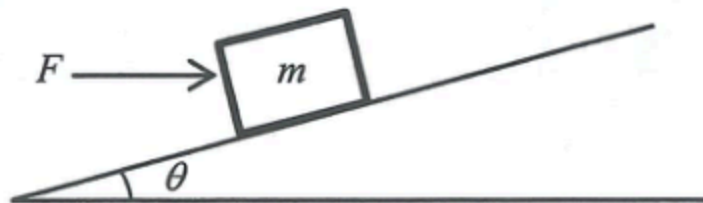
C.



D.



5.



In the above figure, a horizontal force F is applied to a block of mass m so as to keep it at rest on a smooth incline making an angle θ with the horizontal. Find the magnitude of F .

- A. $\frac{mg \sin \theta}{\cos \theta}$
- B. $mg \sin \theta \cos \theta$
- C. $\frac{mg \cos \theta}{\sin \theta}$
- D. $mg \sin \theta$

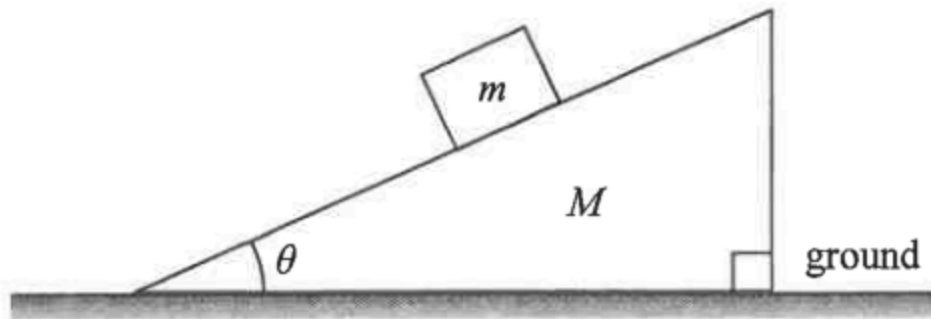
8. An object falls from P to Q as shown below. Throughout the motion, air resistance increases with the speed of the object.



Which of the following descriptions is/are correct ?

- (1) The net force acting on the object is constant throughout the motion.
 - (2) The magnitude of the object's acceleration decreases from P to Q .
 - (3) The kinetic energy gained by the object from P to Q is equal to its loss in gravitational potential energy.
- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

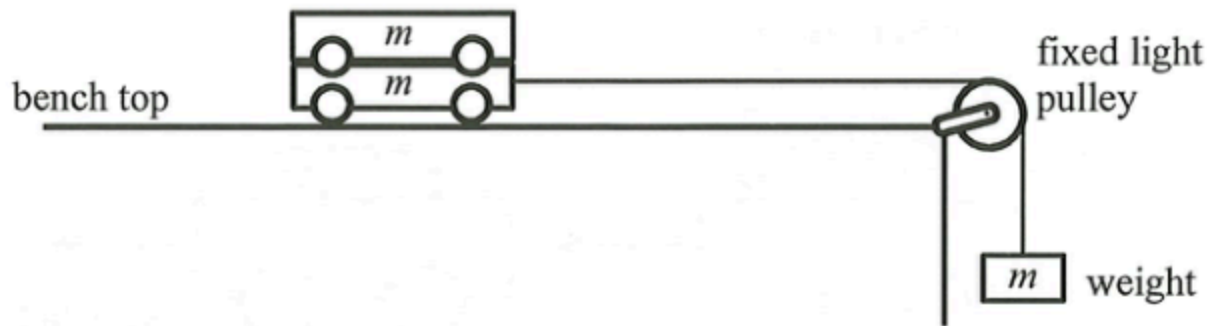
6.



A block of mass m is placed on a wedge of mass M as shown. The system remains at rest. What is the normal force exerted on the wedge by the ground ?

- A. Mg
- B. $(M + m)g$
- C. $Mg + mg \cos \theta$
- D. $Mg + mg \tan \theta$

9. The figure shows a set-up in which a weight of mass m is attached to two trolleys stacked together, each trolley of mass m , via a light inextensible string.

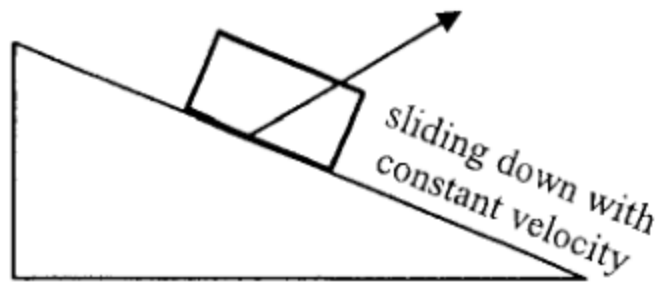


After the weight is released from rest, which of the following is correct? Assume that all contact surfaces are smooth and neglect air resistance. g is the acceleration due to gravity.

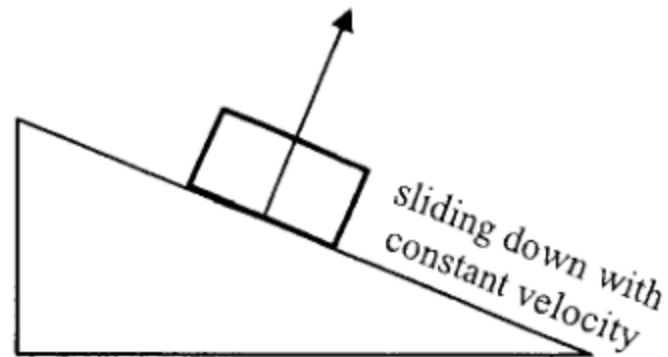
	magnitude of the acceleration	tension in the string
A.	$g/2$	smaller than mg
B.	$g/2$	equal to mg
C.	$g/3$	smaller than mg
D.	$g/3$	equal to mg

5. A block is sliding down a rough incline with constant velocity as shown. Which arrow indicates the direction of **the resultant force acting on the block by the incline** ? Neglect air resistance.

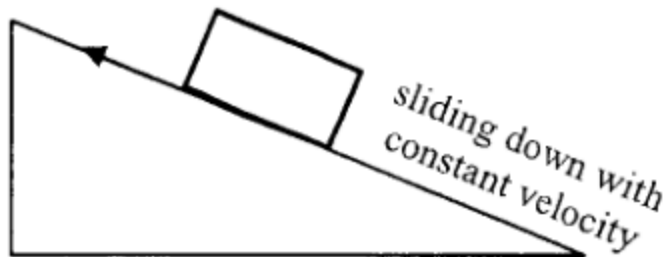
A.



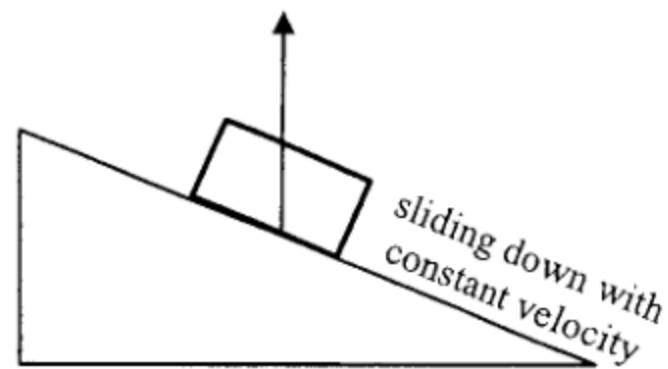
B.



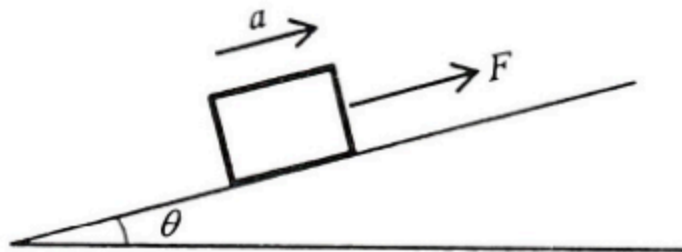
C.



D.

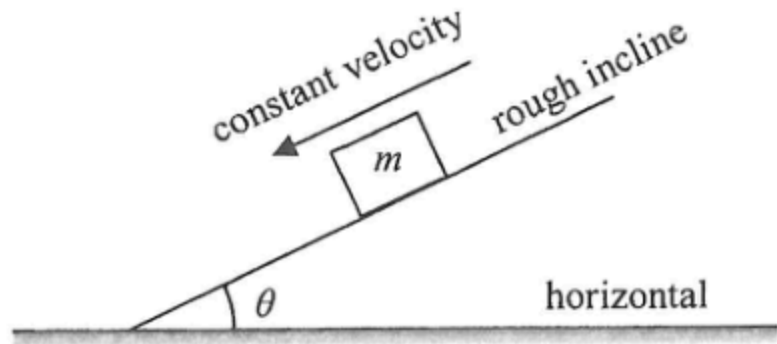


8. A block of mass m is placed on a smooth incline making an angle θ with the horizontal as shown. When a force of magnitude F parallel to the incline is applied to the block, it travels up the incline with an acceleration a . If the applied force becomes $2F$, what would the magnitude of the acceleration be?



- A. greater than $2a$
- B. equal to $2a$
- C. between a and $2a$
- D. whether it is greater than $2a$, equal to $2a$ or between a and $2a$ depends on the value of θ

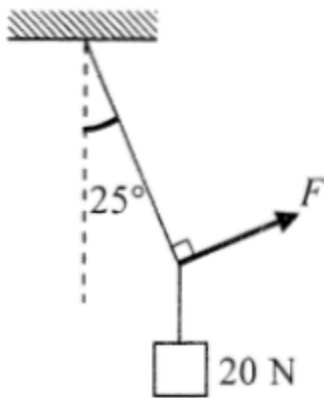
5.



A block of mass m slides down along a fixed rough incline with constant velocity as shown. The incline makes an angle θ with the horizontal. What is the magnitude of the force exerted on the incline by the block ?

- A. zero
- B. mg
- C. $mg \sin \theta$
- D. $mg \cos \theta$

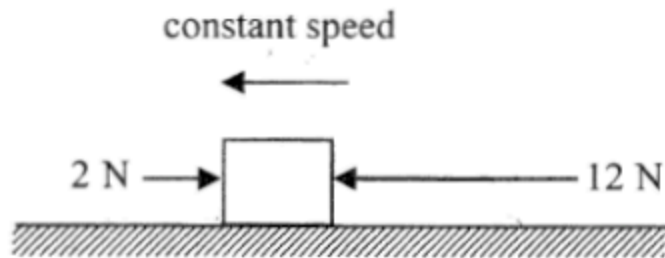
5.



A block of weight 20 N is suspended by a light string from the ceiling. A force F is applied such that the block is displaced to one side with the string making an angle of 25° with the vertical as shown. Find the magnitude of F .

- A. 8.5 N
- B. 9.3 N
- C. 18.1 N
- D. 47.3 N

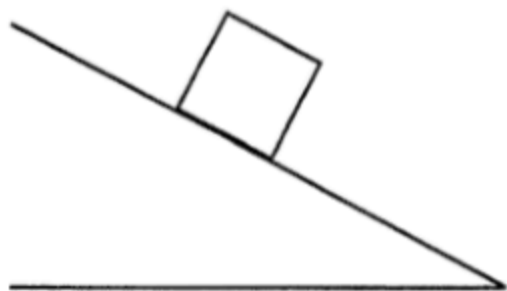
7.



A block on a rough horizontal surface is moving to the left with constant speed under two horizontal forces 2 N and 12 N indicated as shown. If the force of 12 N is suddenly removed, what is the net force acting on the block at that instant ?

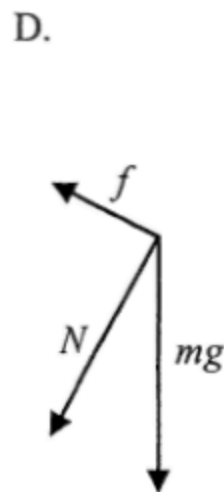
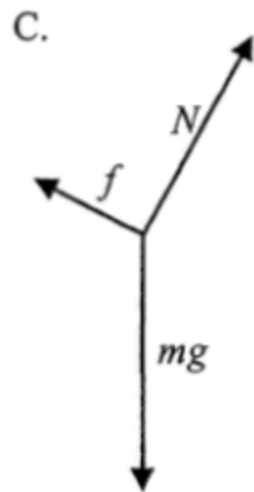
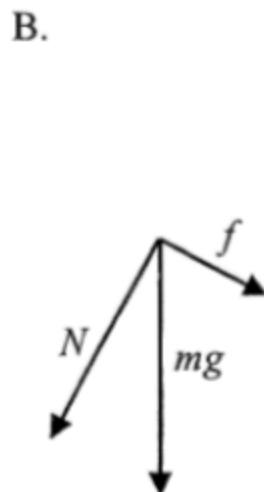
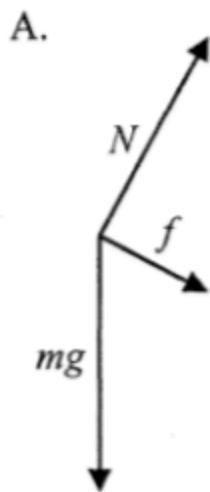
- A. 12 N
- B. 10 N
- C. 8 N
- D. 2 N

7. A block of mass m stays at rest on a rough inclined plane as shown.

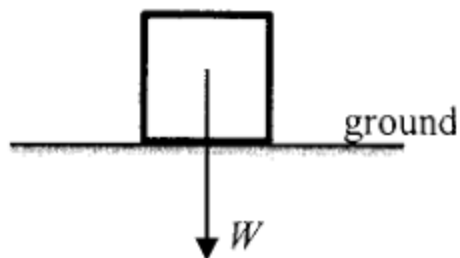


Which of the following diagrams correctly shows the forces acting on the block ?

(N is the normal reaction from the inclined plane, and f is the frictional force between the block and the plane.)



4. A block of weight W is at rest on a horizontal ground as shown.

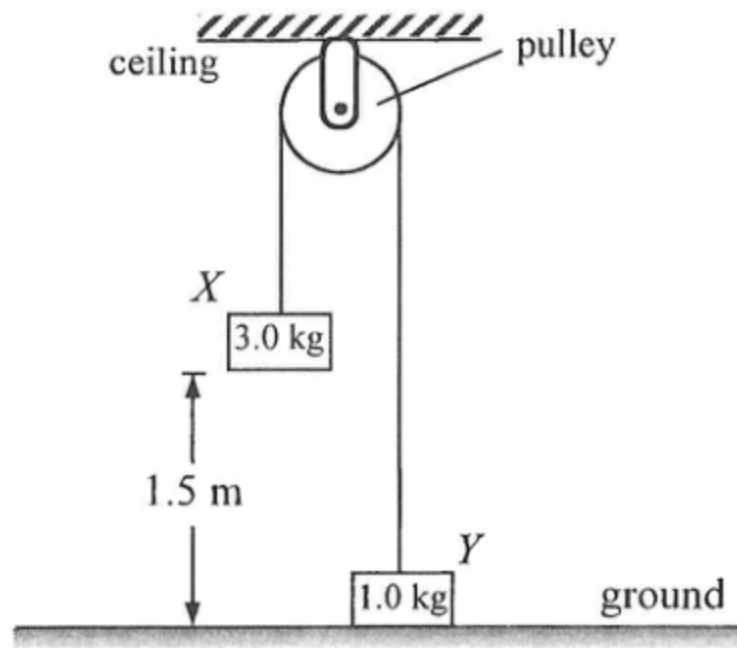


The force acting on the block by the ground is R . Which of the following statements is/are correct ?

- (1) R and W are opposite in direction.
- (2) R and W are equal in magnitude.
- (3) R and W is an action-and-reaction pair.

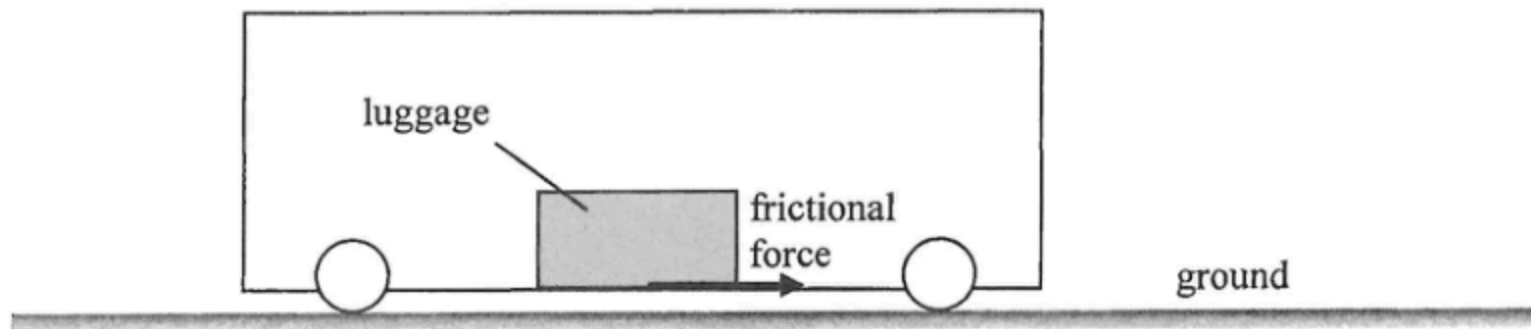
- A. (1) only
- B. (1) and (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

8. Block X and Y of mass 3.0 kg and 1.0 kg respectively are connected by a light inextensible string passing over a smooth light pulley fixed to the ceiling as shown. The blocks are released from rest. Find the time taken for block X to reach the ground. Neglect air resistance.



- A. 0.55 s
- B. 0.64 s
- C. 0.68 s
- D. 0.78 s

5. A train is moving horizontally along a straight line. A piece of luggage rests on the floor of the train. The diagram shows the frictional force acting on the luggage, directed to the right. Neglect air resistance.

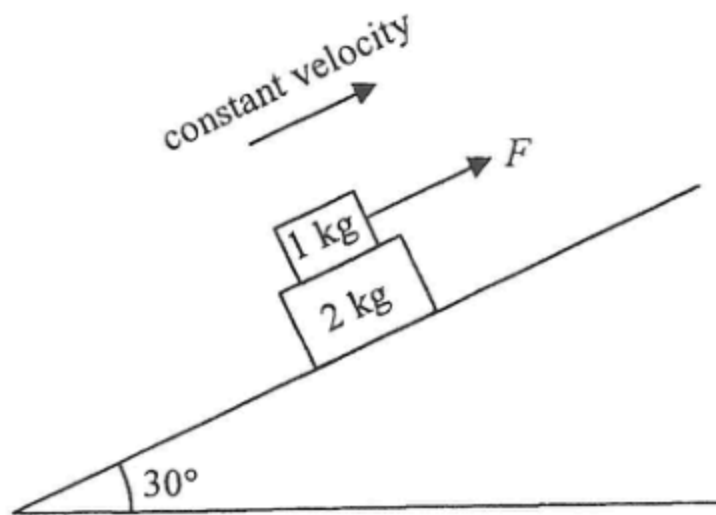


Observing from the ground, which of the following statements must be correct ?

- (1) The train is travelling to the right.
- (2) The train is accelerating.
- (3) The direction of acceleration of the luggage is to the right.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

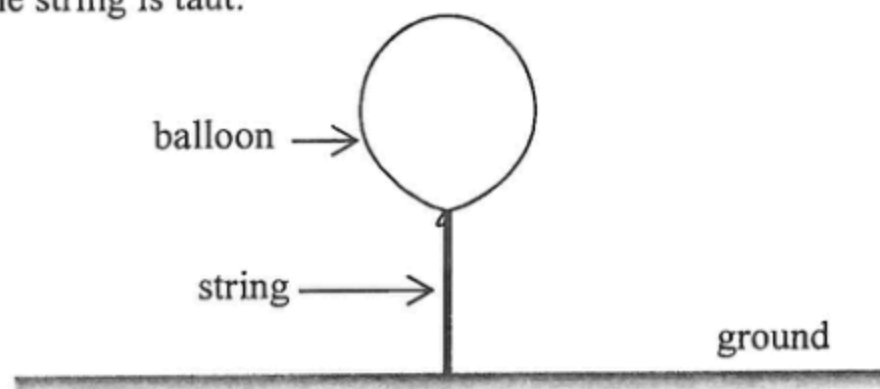
6.



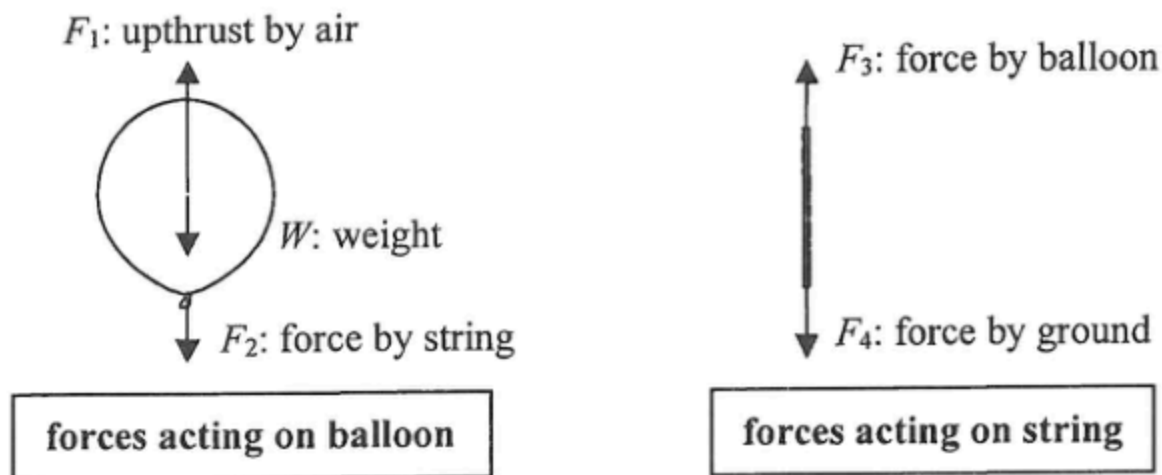
A 1-kg block is placed on a 2-kg block, and both are on a smooth incline at 30° to the horizontal. A constant force F is applied to the 1-kg block upward along the incline so that the two blocks move up the incline together with the same constant velocity as shown. What is the frictional force acting on the 1-kg block by the 2-kg block? Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

- A. 4.91 N upward along the incline
- B. 9.81 N upward along the incline
- C. 4.91 N downward along the incline
- D. 9.81 N downward along the incline

8. One end of a light and inextensible string is connected to a floating balloon and the other end is fixed to the ground as shown. The string is taut.



The free-body diagrams of the balloon and the string are shown below.

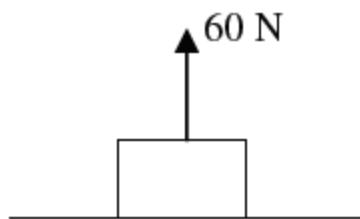


Which of the following is/are action and reaction pair(s) ?

- (1) F_1 and W
- (2) F_2 and F_3
- (3) F_3 and F_4

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

8.

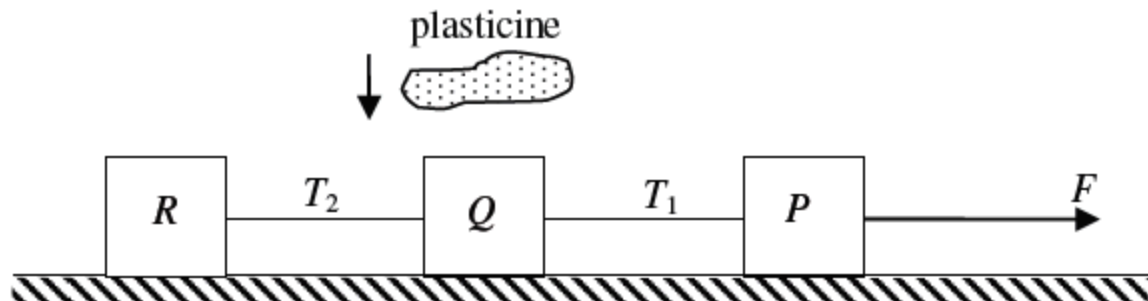


A block of weight 100 N is placed on a horizontal table and a vertical force of 60 N is exerted on the block as shown in the figure above. Which of the following statements is/are correct ?

- (1) The weight of the block is balanced by the force exerted on the block by the table.
- (2) The weight of the block and the force exerted on the table by the block are equal in magnitude.
- (3) The force exerted on the table by the block and the force exerted on the block by the table are an action-reaction pair.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

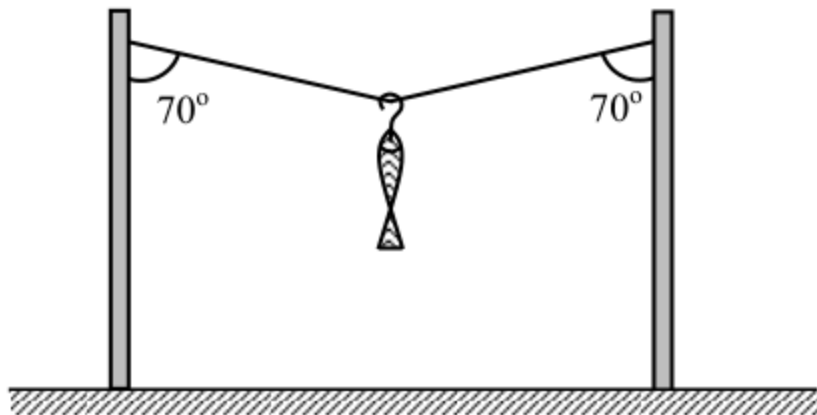
9. Blocks, P , Q and R , connected by light inextensible threads, are placed on a smooth horizontal surface as shown. A constant force F is applied to P so that the whole system travels to the right with acceleration.



A lump of plasticine is placed on Q and it moves together with Q . If the applied force F remains unchanged, how would the tensions T_1 and T_2 in the two threads change ?

	Tension T_1	Tension T_2
A.	increase	decrease
B.	increase	increase
C.	decrease	decrease
D.	decrease	increase

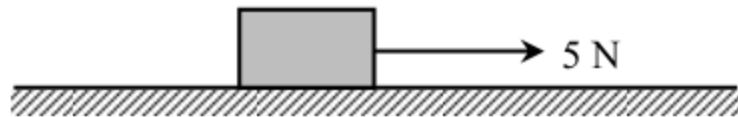
5.



A fish is hung on a light string as shown above. The tension in the string is 10 N. Find the total weight of the fish and the hook.

- A. $20 \sin 70^\circ \text{ N}$
- B. $20 \cos 70^\circ \text{ N}$
- C. $10 \sin 70^\circ \text{ N}$
- D. $10 \cos 70^\circ \text{ N}$

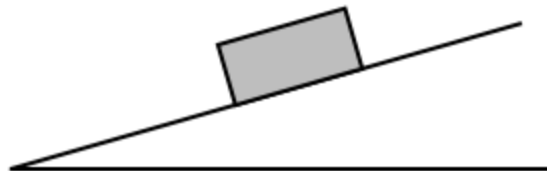
6.



A 1 kg block is pulled by a horizontal force of 5 N and moves with an acceleration of 2 m s^{-2} on a rough horizontal plane. Find the frictional force acting on the block.

- A. zero
- B. 2 N
- C. 3 N
- D. 7 N

8.



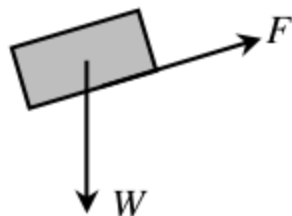
A block remains at rest on a rough inclined plane. Which diagram shows all the forces acting on the block ?

Note : W = gravitational force acting on the block,

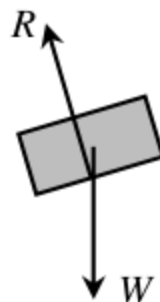
R = normal reaction exerted by the inclined plane on the block, and

F = friction acting on the block.

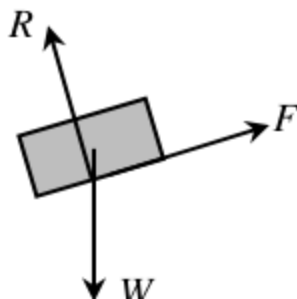
A.



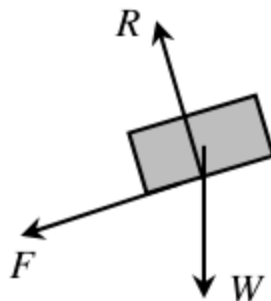
B.



C.



D.



9. Kelvin is standing on a balance inside a lift. The table shows the readings of the balance in three situations.

Motion of the lift	Reading of the balance
moving upwards with a uniform speed	R_1
moving downwards with a uniform speed	R_2
moving upwards with an acceleration	R_3

Which relationship is correct ?

- A. $R_1 = R_2 > R_3$
- B. $R_3 > R_1 = R_2$
- C. $R_1 > R_2 > R_3$
- D. $R_3 > R_1 > R_2$

11. Which of the following pairs of forces is/are example(s) of action and reaction ?

- (1) The centripetal force keeping a satellite in orbit round the earth and the weight of the satellite.
- (2) The air resistance acting on an object falling through the air with terminal velocity and the weight of the object.
- (3) The forces of attraction experienced by two parallel wires carrying currents in the same direction.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only